

LAB 1 — USE-CASE FLOW SPECIFICATION

UC-01: Apply for Micro-Loan

Primary Actor	Borrower
Supporting Actors	Peer Lender; Notification Service
Preconditions	Borrower is registered and authenticated; required financial information is available; a valid loan product/term is available.
Postconditions	Application is stored with a risk tier. If approved, a structured repayment schedule is generated and the loan becomes eligible for peer funding; if rejected, the application is not offered for funding.

Main Success Scenario

1. Borrower selects "Apply for Micro-Loan."
2. System displays the loan application form and required financial fields.
3. Borrower enters loan amount, term, income, outstanding debt, repayment history, and employment stability information.
4. System validates that all mandatory financial data is complete and in an acceptable format.
5. System computes the borrower's credit risk rating using the configured heuristic rules.
6. System displays the calculated risk tier and continues the eligibility decision.
7. System approves the application when the borrower satisfies the configured risk/eligibility rules.
8. System creates the loan record and calculates a structured repayment schedule from the approved loan terms.
9. System stores the schedule and makes the approved loan available to peer lenders.
10. System confirms successful submission to the borrower.
11. Use case ends successfully.

Alternate Flow — Incomplete Financial Profile / High Risk

- 4a. If mandatory financial information is missing or invalid, the system identifies the missing/invalid fields and prevents risk assessment.
 - 4a1. Borrower corrects the information and resubmits; the flow returns to Step 4.
- 5a. If the calculated risk tier is High and the configured eligibility rule rejects the application, the system records the application as rejected.
 - 5a1. The system does not generate a funding opportunity or repayment schedule and informs the borrower that the application was not approved.

Traceability: This flow directly exercises FR-001 (risk assessment) and FR-003 (repayment schedule generation), while the rejection path demonstrates a testable exception to the successful application flow.